



2023 YEAR 12 ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen.
- Calculators approved by NESA may be used.
- Answer all questions in the booklet provided.
- In Questions 6 – 8, show relevant mathematical reasoning and/or calculations.
- Relevant formulas are provided in the answer booklet

Total marks: 38

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section.

Section II – 33 marks (pages 4 – 8)

- Attempt Questions 6 – 8
- Allow about 52 minutes for this section.

Section I

5 marks

Attempt questions 1 - 5

Allow about 8 minutes for this section

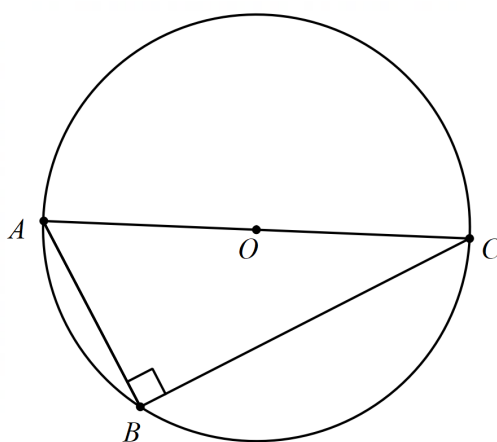
Use the multiple-choice answer page in your booklet to answer Questions 1 – 5.

- 1) If a vector $\underline{u}$ is multiplied by a scalar λ , which of the following expressions represents its magnitude?
- A) $|\underline{u}| + \lambda$
- B) $|\underline{u}|^2 + \lambda^2$
- C) $\underline{u}|\lambda|$
- D) $|\underline{u}|\lambda$
- 2) If $\underline{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -4 \\ -8 \end{pmatrix}$ then which of the following is the length of the projection of $\underline{b}$ onto $\underline{a}$?
- A) $\frac{16}{\sqrt{5}}$
- B) $-\frac{16}{\sqrt{5}}$
- C) $\frac{4}{\sqrt{5}}$
- D) $-\frac{4}{\sqrt{5}}$
- 3) A force of magnitude 18 newtons acts on a body at an angle of 150° in the anticlockwise direction to the vector $\underline{i}$. Which of the following is the vector representation of this force?
- A) $-9\underline{i} + 9\sqrt{3}\underline{j}$
- B) $-9\sqrt{3}\underline{i} + 9\underline{j}$
- C) $-9\underline{i} - 9\sqrt{3}\underline{j}$
- D) $9\sqrt{3}\underline{i} - 9\underline{j}$

4) Vectors $\underline{p}$ and $\underline{q}$ are such that $|\underline{p}| = 3$ and $|\underline{q}| = 4$ and $\underline{p} \cdot \underline{q} = 10$. What is the value of $\underline{q} \cdot (\underline{p} + \underline{q})$?

- A) 0
- B) 14
- C) 26
- D) 28

5) A circle with centre O has radius $\overrightarrow{OA} = \underline{a}$, the points B and C are on the circle such that $\overrightarrow{BC} = \underline{b}$.



Which one of the following statements must be true?

- A) $\underline{a} = \frac{1}{2}\underline{b}$
- B) $\underline{a} = -\frac{1}{2}\underline{b}$
- C) $2\underline{a} \cdot \underline{b} = -\underline{b} \cdot \underline{b}$
- D) $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{b}$

End of Section I

Section II (33 marks)

Attempt Questions 6 - 8

Allow about 52 minutes for this section

Answer each question on the appropriate pages of your answer booklet. Extra paper is available.
Your responses should include relevant calculations and/or mathematical reasoning.

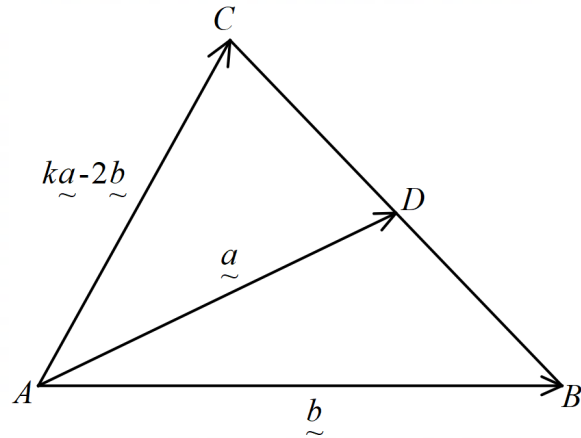
Question 6 (12 marks)

- a) Given that $OACB$ is a parallelogram, where $\overrightarrow{OA} = \underline{a} = -2\underline{i} + 4\underline{j}$ and $\overrightarrow{OB} = \underline{b} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Find;
- i) $3\underline{b} - 2\underline{a}$ 1
 - ii) the magnitude of $\overrightarrow{OA}$ 1
 - iii) the position vector $\overrightarrow{OC}$ 1
 - iv) $\underline{a} \cdot \underline{b}$ 1
 - v) the angle θ between vectors $\underline{a}$ and $\underline{b}$ correct to the nearest degree. 1
- b) Evaluate p and q if $4\begin{pmatrix} p \\ -2 \end{pmatrix} - 3\begin{pmatrix} 2 \\ q \end{pmatrix} = \begin{pmatrix} 14 \\ -8 \end{pmatrix}$. 2
- c) Consider the vector $\underline{a} = 2\underline{i} + 2\sqrt{3}\underline{j}$.
- i) Find the unit vector in the direction of $\underline{a}$. 1
 - ii) Given that vector $\underline{b} = m\underline{i} - 2\underline{j}$ is perpendicular to $\underline{a}$, find the value of m . 1
- d) Use the method of mathematical induction to prove that $5^{2n} + 5^n + 2$ is divisible by 4, where n is positive integer. 3

End of Question 6

Question 7 (9 marks)

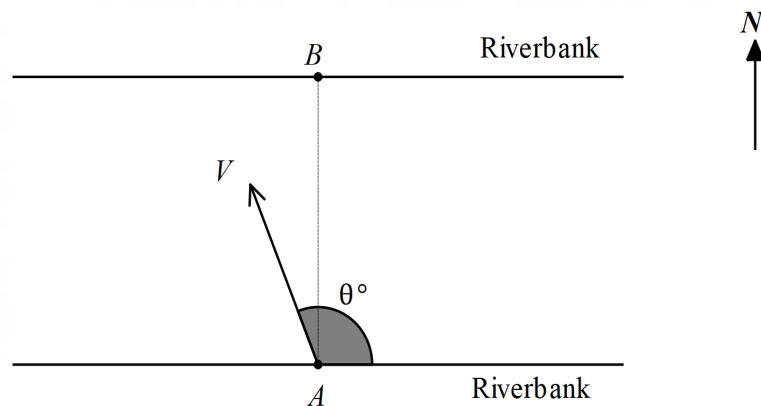
- a) ABC is a triangle where $\overrightarrow{AC} = k\vec{a} - 2\vec{b}$, $\overrightarrow{AD} = \vec{a}$ and $\overrightarrow{AB} = \vec{b}$. Find the value of k , if point D lies on side BC of a triangle. **3**



- b) Consider the vectors $\vec{u} = -3\vec{i} + 4\vec{j}$ and $\vec{v} = 2\vec{i} - 2\vec{j}$.
- i) Find the vector projection of $\vec{u}$ onto $\vec{v}$. **2**
- ii) Find a possible vector $\vec{w}$ such that $|\text{proj}_{\vec{u}} \vec{w}| = 2$. **1**

Question 7 continues on next page

- c) Andy wants to swim across the river from position A to position B shown on the diagram below. The riverbanks are parallel straight lines, and $\overline{AB}$ is perpendicular to the riverbanks. The speed of the river flow is 0.5 m/s from west to east and the river is 0.7 km wide. Assume Andy swims at a constant speed of 1.4 m/s in still water.

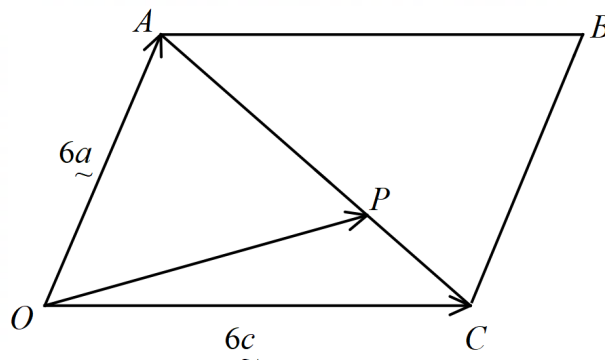


- i) Find the angle θ at which Andy should swim to cross the river in the shortest time from point A to point B . Draw a vector diagram to demonstrate the velocities and show all necessary working out. Give your answer in degrees correct to two decimal places. 2
- ii) Calculate the time Andy would take to travel from point A to point B . Give your answer to the nearest second. 1

End of Question 7

Question 8 (12 marks)

- a) $OABC$ is a parallelogram, where $\overrightarrow{OA} = 6\underset{\sim}{a}$, $\overrightarrow{OC} = 6\underset{\sim}{c}$ and point P is on AC such that $AP:PC = 2:1$.

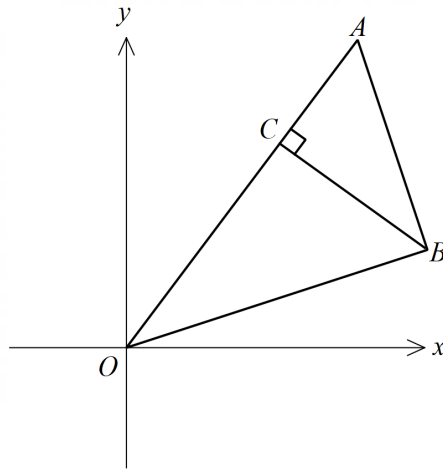


- i) Find the vector $\overrightarrow{OP}$. Give your answer in terms of $\underset{\sim}{a}$ and $\underset{\sim}{c}$. 2
- ii) Given that the midpoint of CB is M , prove that OPM is a straight line. 2
- b) If $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ are unit vectors such that $\underset{\sim}{a} + \underset{\sim}{b} + \underset{\sim}{c} = \underset{\sim}{0}$ then find the value of $\underset{\sim}{a} \cdot \underset{\sim}{b} + \underset{\sim}{b} \cdot \underset{\sim}{c} + \underset{\sim}{c} \cdot \underset{\sim}{a}$. 2
- c) Use the principle of mathematical induction to show that for all integers $n \geq 1$, 3

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{n+2}{n(n+1)2^n} = 1 - \frac{1}{(n+1)2^n}$$

Question 8 continues on next page

- d) The diagram below shows $\triangle OAB$ where $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.



Point C lies on OA such that CB is perpendicular to OA .

i) Show that $\overrightarrow{BC} = \frac{1}{\underline{a} \cdot \underline{a}} [(\underline{a} \cdot \underline{b})\underline{a} - (\underline{a} \cdot \underline{a})\underline{b}]$.

2

ii) Hence, show that the area of $\triangle OAB$ is $\frac{1}{2|\underline{a}|} |(\underline{a} \cdot \underline{b})\underline{a} - (\underline{a} \cdot \underline{a})\underline{b}|$

1

End of Exam

2023 Year 12 Assessment Task 1

Solutions

Section 1 Multiple Choice

1 D

2 A

3 B

4 C

5 C

1 Magnitude = $|\underline{u}|$

(D)

2 length of projection of $\underline{b}$ onto $\underline{a} = \left| \frac{\underline{b} \cdot \underline{a}}{\underline{a} \cdot \underline{a}} \right|$

$$= \left| \frac{2(-4) + 1(-8)}{\sqrt{2^2 + 1^2}} \right|$$
$$= \left| \frac{-8 - 8}{\sqrt{5}} \right|$$
$$= \frac{16}{\sqrt{5}}$$

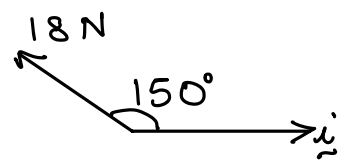
(A)

3 Vector representation of force is:

$$(18 \cos 150^\circ) \underline{i} + (18 \sin 150^\circ) \underline{j}$$

$$18 \left(-\frac{\sqrt{3}}{2} \right) \underline{i} + 18 \left(\frac{1}{2} \right) \underline{j}$$

$$-9\sqrt{3} \underline{i} + 9 \underline{j}$$



(B)

4 $\underline{q} \cdot (\underline{p} + \underline{q}) = \underline{q} \cdot \underline{p} + \underline{q} \cdot \underline{q}$

$$= 10 + 16$$
$$= 26$$

(C)

5

$$\vec{OA} = \underline{a}, \quad \vec{BC} = \underline{b}$$

$$\vec{BA} = \vec{BC} + \vec{CA}$$

$$= \underline{b} + 2\underline{a} \quad (\because \vec{CA} = 2\vec{OA})$$

$$\vec{BA} \perp \vec{BC}$$

$$\therefore \vec{BA} \cdot \vec{BC} = 0$$

$$(\underline{b} + 2\underline{a}) \cdot \underline{b} = 0$$

$$\underline{b} \cdot \underline{b} + 2\underline{a} \cdot \underline{b} = 0$$

$$2\underline{a} \cdot \underline{b} = -\underline{b} \cdot \underline{b}$$

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END OF Multiple Choice Section.

Section-2

Question-6 a)

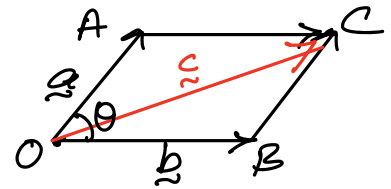
$$\begin{aligned} \text{i) } 3\mathbf{b} - 2\mathbf{a} &= 3(3\mathbf{i} + 5\mathbf{j}) - 2(-2\mathbf{i} + 4\mathbf{j}) \\ &= 13\mathbf{i} + 7\mathbf{j} \end{aligned}$$

1 mark correct answer

$$\text{ii) } |\vec{OA}| = \sqrt{(-2)^2 + 4^2} = \sqrt{20} = 2\sqrt{5}$$

1 mark correct answer

$$\begin{aligned} \text{iii) } \vec{OC} &= \vec{OA} + \vec{AC} \quad \text{or} \quad \vec{OB} + \vec{BC} \\ &= \vec{OA} + \vec{OB} \\ &= (-2\mathbf{i} + 4\mathbf{j}) + (3\mathbf{i} + 5\mathbf{j}) \\ &= \mathbf{i} + 9\mathbf{j} \end{aligned}$$



1 mark correct answer

$$\begin{aligned} \text{iv) } \mathbf{a} \cdot \mathbf{b} &= (-2)(3) + (4)(5) \\ &= -6 + 20 = 14 \end{aligned}$$

1 mark correct answer

$$\begin{aligned} \text{v) } \cos \theta &= \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \\ &= \frac{14}{\sqrt{20} \sqrt{34}} \end{aligned}$$

$$\theta = 57.5288$$

$$\theta = 58^\circ$$

1 mark correct answer

Question-6

$$b) \quad 4 \begin{pmatrix} p \\ -2 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ q \end{pmatrix} = \begin{pmatrix} 14 \\ -8 \end{pmatrix}$$

$$\begin{pmatrix} 4p-6 \\ -8-3q \end{pmatrix} = \begin{pmatrix} 14 \\ -8 \end{pmatrix}$$

$$4p-6=14 \quad \& \quad -8-3q=-8$$

$$p=5$$

$$q=0$$

2 marks correct solution

1 mark for correct value of either p or q.

(OR)

Correct process to multiply and equating two sides of the vector to perform equations.

c)

$$i) \quad \underline{a} = 2\underline{i} + 2\sqrt{3}\underline{j}$$

$$|\underline{a}| = \sqrt{(2)^2 + (2\sqrt{3})^2} = 4$$

$$\underline{\hat{a}} = \frac{2\underline{i} + 2\sqrt{3}\underline{j}}{4} = \frac{1}{2}(\underline{i} + \sqrt{3}\underline{j})$$

or $\frac{\underline{i} + \sqrt{3}\underline{j}}{2}$

1 mark correct solution

$$ii) \quad \text{Since } \underline{a} \perp \underline{b} \therefore \underline{a} \cdot \underline{b} = 0$$

$$2m - 4\sqrt{3} = 0$$

$$m = 2\sqrt{3}$$

1 mark correct solution

d) Let $S(n)$ be the statement that $5^{2n} + 5^n + 2$ is divisible by 4.

Show $S(1)$ is true

$$5^2 + 5 + 2 = 25 + 5 + 2 = 32, \text{ which is divisible by 4.}$$

$\therefore S(1)$ is true.

Assume $S(k)$ is true

$$\text{i.e. } 5^{2k} + 5^k + 2 = 4M, \text{ for some integer } M.$$

Show $S(k+1)$ is true.

$$\text{RTP: } 5^{2(k+1)} + 5^{k+1} + 2 = 4N, \text{ for some integer } N.$$

$$\text{L.H.S} = 5^{2(k+1)} + 5^{k+1} + 2$$

$$= 25(5^{2k}) + 5(5^k) + 2$$

$$= 25(4M - 5^k - 2) + 5(5^k) + 2$$

(using assumption)

$$= 100M - 20(5^k) - 48$$

$$= 4(25M - 5(5^k) - 12)$$

$$= 4N \quad \text{where } N = 25M - 5^{k+1} - 12 \text{ is integer.}$$

$$= \text{R.H.S}$$

$\therefore S(k+1)$ is true if $S(k)$ is assumed true.

Since $S(1)$ is true, by the principle of mathematical induction, $S(n)$ is true for all positive integers n .

3 marks correct induction.
2 marks to use the assumption to prove the required result.
1 mark correct base case.

Question-7 a)

Since D lies on BC $\therefore \vec{BC} \parallel \vec{BD}$

$\therefore \vec{BC} = \lambda \vec{BD}$ for some real λ .

$$\vec{AC} - \vec{AB} = \lambda(\vec{AD} - \vec{AB})$$

$$k\vec{a} - 2\vec{b} - \vec{b} = \lambda(\vec{a} - \vec{b})$$

$$k\vec{a} - 3\vec{b} = \lambda\vec{a} - \lambda\vec{b}$$

Coeff. of $\vec{b} \Rightarrow \lambda = 3$

& Coeff. of $\vec{a} \Rightarrow k = \lambda = 3$.

3 marks correct solution.

2 marks for correctly finds two equations from $\vec{BC}$, $\vec{BD}$ or $\vec{DC}$.

1 mark to find $\vec{BC}$ or $\vec{BD}$ or realising C, D, B are collinear points.

b) i) $\vec{u} = -3\vec{i} + 4\vec{j}$, $\vec{v} = 2\vec{i} - 2\vec{j}$

$$\vec{u} \cdot \vec{v} = -6 - 8 = -14 \quad \vec{v} \cdot \vec{v} = 4 + 4 = 8$$

$$\begin{aligned} \text{proj}_{\vec{v}} \vec{u} &= \left(\frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \vec{v} \\ &= -\frac{14}{8} (2\vec{i} - 2\vec{j}) \\ &= -\frac{7}{4} (2\vec{i} - 2\vec{j}) \\ &= -\frac{7}{2} (\vec{i} - \vec{j}) \end{aligned}$$

2 marks correct solution

1 mark to attempt the correct process to find the projection.

ii) Let $\vec{w} = x\vec{i} + y\vec{j}$

$$\text{scal}_{\vec{u}} \vec{w} = \frac{\vec{w} \cdot \vec{u}}{|\vec{u}|} = -\frac{3x + 4y}{5}$$

Since $\text{scal}_{\vec{u}} \vec{w} = 2$

$$\therefore -3x + 4y = 10$$

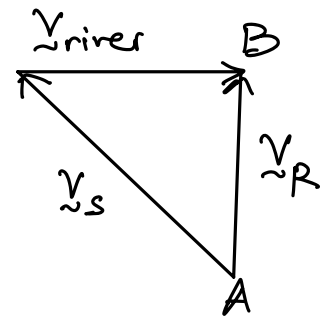
1 mark correct answer

$\therefore \vec{w} = -2\vec{i} + \vec{j}$ or any possible set of x & y that satisfies the equation.

c) River flow velocity vector $\underline{V}_{\text{river}}$

Let swimmers velocity vector $\underline{V}_s$

and Resultant velocity vector $\underline{V}_R$



$$\therefore \underline{V}_s + \underline{V}_{\text{river}} = \underline{V}_R \quad \text{--- (1)}$$

R is the speed in m/s.

$$\underline{V}_s = \begin{pmatrix} 1.4 \cos \theta \\ 1.4 \sin \theta \end{pmatrix} \quad \underline{V}_{\text{river}} = \begin{pmatrix} 0.5 \cos 0^\circ \\ 0.5 \sin 0^\circ \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix}$$

$$\underline{V}_R = \begin{pmatrix} 0 \\ R \end{pmatrix}$$

Using (1); $\begin{pmatrix} 0 \\ R \end{pmatrix} = \begin{pmatrix} 1.4 \cos \theta \\ 1.4 \sin \theta \end{pmatrix} + \begin{pmatrix} 0.5 \\ 0 \end{pmatrix}$

$$1.4 \cos \theta + 0.5 = 0 \quad \text{--- (2)}$$

$$R = 1.4 \sin \theta \quad \text{--- (3)}$$

from (2); $\cos \theta = \frac{-0.5}{1.4}$
 $\theta = 110.924^\circ$
 $= 110.92^\circ$

2 marks correct solution
1 mark attempt to
create vector equations

$\therefore$ Swimmer should swim at an angle of $110^\circ 55'$ on the South bank at point A.

ii) from (3); $R = 1.4 \sin 110^\circ 55' = 1.307669... \text{ m/s}$

$\therefore$ Time taken to cross river and reach point B is

$$\frac{700}{1.307669...} \approx 535.3033 \text{ seconds}$$

8 mins 55 seconds

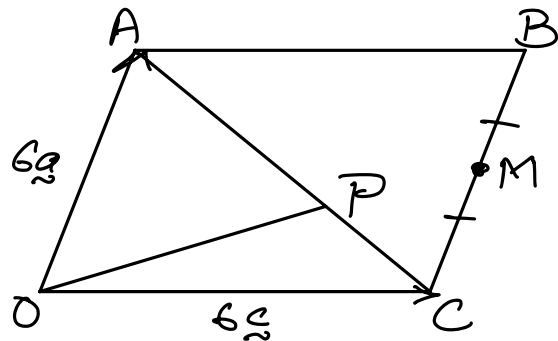
1 mark correct
solution

Question-8

a) i) $AP : PC = 2 : 1$

$$\begin{aligned}\therefore \vec{AP} &= \frac{2}{3} \vec{AC} \\ &= \frac{2}{3} (6\vec{c} - 6\vec{a}) \\ &= 4(\vec{c} - \vec{a})\end{aligned}$$

$$\begin{aligned}\vec{OP} &= \vec{OA} + \vec{AP} \\ &= 6\vec{a} + 4(\vec{c} - \vec{a}) \\ &= 2\vec{a} + 4\vec{c}\end{aligned}$$



{ 2 marks correct solution
1 mark for correct $\vec{AP}$ or $\vec{PC}$.

ii) $\vec{CM} = \frac{1}{2} \vec{CB} = \frac{1}{2} \vec{OA} = \frac{1}{2} (6\vec{a})$
 $= 3\vec{a}$

$$\begin{aligned}\vec{OM} &= \vec{OC} + \vec{CM} \\ &= 6\vec{c} + 3\vec{a} \\ &= 3(2\vec{c} + \vec{a}) \\ &= 3(\vec{a} + 2\vec{c}) \\ &= \frac{3}{2} \vec{OP}\end{aligned}$$

$$\therefore \vec{OM} \parallel \vec{OP}$$

Since they both share the common point O.

$\therefore$ OPM is a straight line.

{ 2 marks correct proof
1 mark to find $\vec{CM}$
or $\vec{OM}$.

b) Since $\underline{a}$, $\underline{b}$ and $\underline{c}$ are unit vectors

$$\therefore |\underline{a}| = |\underline{b}| = |\underline{c}| = 1$$

$$(\underline{a} + \underline{b} + \underline{c}) \cdot (\underline{a} + \underline{b} + \underline{c}) = 0$$

$$\therefore \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{a} \cdot \underline{c} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} + \underline{b} \cdot \underline{c} + \underline{c} \cdot \underline{a} + \underline{c} \cdot \underline{b} + \underline{c} \cdot \underline{c} = 0$$

$$2(\underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{c} + \underline{c} \cdot \underline{a}) + |\underline{a}|^2 + |\underline{b}|^2 + |\underline{c}|^2 = 0$$

$$2(\underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{c} + \underline{c} \cdot \underline{a}) + 1 + 1 + 1 = 0$$

$$\therefore \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{c} + \underline{c} \cdot \underline{a} = -\frac{3}{2}$$

$\left\{ \begin{array}{l} 2 \text{ mark correct solution} \\ 1 \text{ mark attempt to} \\ \text{prove by using unit} \\ \text{vector with } \underline{a} + \underline{b} + \underline{c} = 0 \end{array} \right.$

8c)

Let $S(n)$ be the statement that

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{n+2}{n(n+1)2^n} = 1 - \frac{1}{(n+1)2^n}$$

Show $S(1)$ is true

$$\text{L.H.S} = \frac{1+2}{1(1+1)2^1} = \frac{3}{1 \times 2 \times 2} = \frac{3}{4}$$

$$\text{R.H.S} = 1 - \frac{1}{(1+1)2^1} = 1 - \frac{1}{4} = \frac{3}{4}$$

Since $\text{L.H.S} = \text{R.H.S}$

$\therefore S(1)$ is true.

Assume $S(k)$ is true

$$\text{i.e., } \frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{k+2}{k(k+1)2^k} = 1 - \frac{1}{(k+1)2^k}$$

Show $S(k+1)$ is true

$$\text{i.e., } \frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{k+2}{k(k+1)2^k} + \frac{k+3}{(k+1)(k+2)2^{k+1}} = 1 - \frac{1}{(k+2)2^{k+1}}$$

$$\text{L.H.S} = \underbrace{\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{k+2}{k(k+1)2^k}}_{S(k)} + \frac{k+3}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \frac{1}{(k+1)2^k} + \frac{k+3}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \left(\frac{2(k+2) - (k+3)}{(k+1)(k+2)2^{k+1}} \right)$$

$$= 1 - \left(\frac{2k+4-k-3}{(k+1)(k+2)2^{k+1}} \right)$$

$$= 1 - \frac{(k+1)}{(k+1)(k+2)2^{k+1}}$$

$$= 1 - \frac{1}{(k+2)2^{k+1}}$$

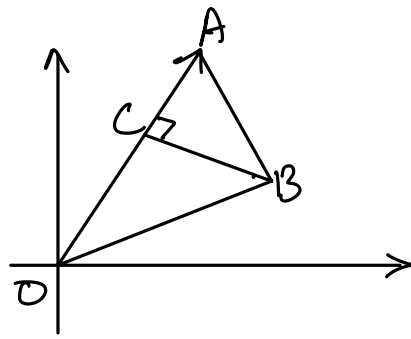
$$= \text{R.H.S}$$

3 marks correct induction.
2 marks to use the assumption to prove the required result.
1 mark correct base case.

$\therefore S(k+1)$ is true if $S(k)$ is assumed true.

Since $S(1)$ is true, by the principle of mathematical induction, $S(n)$ is true for all positive integers n .

8d)



$\vec{OC}$ is the projection of $\vec{OB}$ onto $\vec{OA}$

$$\begin{aligned}\therefore \vec{OC} &= \text{proj}_{\vec{a}} \vec{b} \\ &= \left(\frac{\vec{b} \cdot \vec{a}}{\vec{a} \cdot \vec{a}} \right) \vec{a}\end{aligned}$$

2 marks correct proof
1 mark to find $\vec{OC}$ in terms of $\vec{a}$ and $\vec{b}$.

$$\begin{aligned}\vec{BC} &= \vec{OC} - \vec{OB} \\ &= \left(\frac{\vec{b} \cdot \vec{a}}{\vec{a} \cdot \vec{a}} \right) \vec{a} - \vec{b} \\ &= \left(\frac{\vec{b} \cdot \vec{a}}{\vec{a} \cdot \vec{a}} \right) \vec{a} - \left(\frac{\vec{a} \cdot \vec{a}}{\vec{a} \cdot \vec{a}} \right) \vec{b} \\ &= \frac{1}{\vec{a} \cdot \vec{a}} \left[(\vec{b} \cdot \vec{a}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b} \right] \\ &= \text{Hence shown}\end{aligned}$$

$$\begin{aligned}\text{ii) Area of } \triangle OAB &= \frac{1}{2} |\vec{OA}| |\vec{BC}| \\ &= \frac{1}{2} |\vec{a}| \left| \frac{1}{\vec{a} \cdot \vec{a}} \left[(\vec{b} \cdot \vec{a}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b} \right] \right| \\ &= \frac{|\vec{a}|}{2|\vec{a}|^2} \left| (\vec{b} \cdot \vec{a}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b} \right| \\ &= \frac{1}{2|\vec{a}|} \left| (\vec{b} \cdot \vec{a}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b} \right|\end{aligned}$$

Hence shown.

End of Solutions

1 mark correct proof.